Biological control and functional biodiversity in agricultural production – successful cases to improve food security in Africa

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Abstract: Because of the introduction of two arthropods from South America into Africa in the mid-1970s, cassava production fell precipitously and food security was endangered on a continental scale. Biological control of the cassava mealybug by the introduced wasp *Anagyrus lopezi* (Hym., Encyrtidae) and of the cassava green mite by the predatory mite *Typhlodromalus aripo* (Acari, Phytoseiidae) lowered the mean population level of both pests to acceptable levels. Total savings that accrued to the mostly poor small-holders are estimated at $10 billion (over 27 countries) and 2 billion (over 3 countries), respectively. Differences in the outcome of the two programs, actions needed where and when infestation levels still are above the threshold, as well as non-monetary gains in avoided pollution and biodiversity preservation are discussed.

Key words: cassava pests, biological control agents, integrated control

Cassava was introduced into Africa by the Portuguese in the 16th century and has become the main staple food for about half the population of sub-Saharan Africa. Up to today, almost all cassava is produced without insecticides and fertilizer, because the value of the crop is relatively low. The label 'organic', though correct, is, however, never used. Productivity on subsistence farms, even with new, higher yielding and virus-resistant varieties, is still only about a fourth of the on-station yield. Cassava is processed and detoxified, mostly by women, into a dry, cyanide-free powder, 'gari', which can be stored and easily commercialized.

Up to recently, few indigenous insects and pathogens attacked this crop. In the mid-1970s the situation changed, when two arthropod pests in short succession invaded the continent. Cassava mealybug *Phenacoccus manihoti* Matile-Ferrero (Hom., Pseudococcidae) was observed first in the Congo, then in Nigeria/Benin, Senegal/Gambia and later spread across the continent at the speed of about 150 km per year, leaving a trail of destruction. The mealybug caused cassava buds to curl and the stems to twist, and tuber production fell 10 fold. Because no planting sticks were available for the next season, cassava production was sometimes abandoned over vast areas (Neuenschwander, 2001). Around the same time, cassava green mite, *Mononychellus tanajoa* (Bondar) (Acari, Tetranychidae) was introduced in the area of Lake Victoria and spread across the entire continent, so that today only Madagascar is free of both pests. In response to these invasions by alien arthropods, two interconnected biological control projects with extensive international collaboration between IITA, CIAT, FAO, CAB International, the Inter-African Phytosanitary Council, National development agencies and universities in Europe and America, as well as universities and national programs of over 20 African countries were launched.

Mealybug parasitoids were first imported from northern South America, but did not attack the cassava mealybug in Africa. Their host was later identified as a new mealybug species, and the search started again. The original home of the cassava mealybug was later discovered in southern South America and by 1981 the parasitic wasp *Anagyrus (Apoanagyrus, Epidinocarsis) lopezi* (De Santis) (Hym., Encyrtidae) was introduced to Nigeria – and later to Benin - via quarantine in England.. In quarantine, this exotic species was tested for diseases, for being specific to the mealybug, and for not attacking other insects, particularly parasitoids as a hyperparasitoid. Mealybug predators, mainly coccinellids, followed, but they never became important. Foreign exploration for predators of the cassava green mite also started in Columbia, yielding a plethora of predatory phytoseiid mites, 10 of which were passed through quarantine in the Netherlands and introduced into Africa; none established, however. The search for natural enemies then moved to climatically matched areas in Northeastern Brazil which eventually yielded six species that were introduced into Africa after passing through quarantine in the Netherlands. Initially, two species – *Typhlodromalus manihoti* Moraes and *Neoseiulus idaeus* Denmark & Muma - established in Africa, though their dispersal and impact remained limited and *N. idaeus* later became extinct (Yaninek and Hanna 2003). It was only four years later, when *Typhlodromalus aripo* DeLeon was introduced that the control project took off.

The exotic agents were reared in glass houses and laboratories first in Ibadan, IITA-Nigeria, and since 1988 in Cotonou, IITA-Bénin. National program staffs were trained in rearing and monitoring, and rearing facilities were established in many countries, though most releases were still made with biological control agents out of IITA. On request by national programs, which provided the import permits, the following releases were made in collaboration with national scientists: about 150 releases of *A. lopezi* (sometimes together with two species of coccinellids) between 1981 and 1995 in over 20 countries of sub-Saharan Africa (Neuenschwander, 2001), and 524 releases of three species of predatory mites, mainly *T. aripo*, between 1989 and 2004 in 19 African countries.

Releases were followed up, sometimes over many years, in surveys covering most release countries. Data on the presence and abundance of the exotic agents, their indigenous counterparts, their hosts and other members of the food-webs, as well as on weather, soil conditions, cassava varieties and weeds, cassava yield, and market prices (of cassava and replacement crops) were obtained to quantify establishment, spread, and impact on the pest populations, on plant growth, yields, and the economy. These were complemented by on-station experiments, using different exclusion and other techniques to determine details of impact and of the biology (host finding, reproductive behaviour, migration, alternate hosts, etc.).

Compared to other exotic and indigenous natural enemies, *A. lopezi* proved to be the superior competitor (Neuenschwander, 1996). The female is capable of ovipositing already into 2nd instar hosts, but reserves the bulk of female eggs for the bigger 3rd and 4th instar hosts. Though the host tries to encapsulate and kill the parasitoid larva within its coelom, this immune reaction is overcome by almost all larvae. The adult wasp emerges after only 2 weeks, which is half the generation time of the host, and has an astonishing host finding capacity, based on olfactory cues from mealybug infested plants. Egg laying capacity is rather limited and females have a tendency to leave the host patch immediately, without exploiting it fully. This early dispersal is probably the reason for the relatively slow impact, which makes itself fully felt only about 2 years after release, presumably when immigration matches emigration. The parasitoid reduces host population peaks by about 10 times (depending on the cassava variety), whereas local predators, mainly coccinellids, contribute to a 20% reduction of peak mealybug populations. Since cassava mealybug (at least at low population densities) is essentially monophagous and hardly ever attacked by indigenous parasitoids, and *A. lopezi* does not attack indigenous mealybugs, there are no non-target effects (Neuenschwander and Markham, 2001).

The economic impact of this project (Table 1) has been calculated to be in the order of $10 billion, with a benefit:cost ratio of 1:140 to 1:600, all figures depending on the assumed scenario (loss replacement by more cassava planting, cassava or maize import or food-aid) (Zeddies *et al.*, 2001). This benefit corresponds to the benefit of 20 years of cassava breeding with all the institutional support at international and national levels that this activity commands.

Table 1. Economic impact analysis for two biological control projects in Africa (Neuenschwander, 2004)

Invasive pest species	Cassava mealybug	Cassava green mite
Typical mean, area-wide yield losses	40%	35%
Biological control agent	*Anagyrus lopezi*	*Typhlodromalus aripo*
Start of campaign	1981	1983
Area under economic analysis	27 African nations	Nigeria, Ghana, Benin
Reduction in loss attributed to this pest species	90-95%	80-95%
Estimated savings in US$ million	7971-20226	2157
Source	Zeddies *et al.*, 2001	Alene *et al.*, 2005

The cassava green mite biological control campaign has resulted in a similar continent-wide success, largely through the establishment of Brazilian populations of the predatory mite *T. aripo*. This predator is now established in at least 20 countries in sub-Saharan Africa (Hanna and Toko, 2003; Yaninek and Hanna, 2003). Repeated surveys in at least 12 countries and a total of 18 predator exclusion trials have shown that the predators have reduced cassava green mite densities by 60% and increased cassava root yield up to 80% depending on location and cassava variety (Yaninek and Hanna 2003; Hanna *et al.* 2005), and notably without any negative non-target effects (Zannou *et al.* 2007). For three countries in West Africa where the economic analysis of the benefits have been completed, cassava green mite biological control with *T. aripo* has provided enormous economic benefits at no cost to the farmers: over $2.2 billion from 1983 to 2020 for Benin, Ghana and Nigeria (Table 1; Alene *et al.*, 2005). In the remaining countries where cassava green mite biological control has been achieved similar benefits continue to be generated. Evaluation of the socioeconomic impact of this project in East and Southern Africa is underway.

Parallel to the use of exotic predatory mites, exotic isolates (also from Brazil) of the entomopathogenic fungus *Neozygites tanajoae* Delalibera Jr., Humber & Hajek (Entomophthorales: Neozygitaceae) was released in Benin. For the evaluation, molecular diagnostic tools had to be developed to distinguish African and Brazilian isolates of *N. tanajoae*. The fungus has contributed to further reductions in cassava green mite densities (Hountondji *et al.*, 2002; Hanna and Toko, 2003). It has also been released in two areas in Tanzania, where the exotic predators have, however, not been able to establish and control cassava green mite. Efforts are underway to evaluate the fungus in other countries.

Overall, *A. lopezi* led to complete control in 90-95% of all fields. The rest of the fields, where control was not satisfactory, invariably concerned fields on pure sand without mulch. Under these conditions, the host plant was specially favorable to the mealybug and hostile to the wasp, so that *A. lopezi* was not able to contain its host populations. By contrast, *T. aripo* efficiency proved to depend to a larger extent on environmental (low relative humidity as a limit) and plant factors. Of particular significance are the interactions between cassava clones and *T. aripo*. This predator inhabits the apex (Onzo *et al.*, 2003) of the plant and prefers apices that are 'hairy' over those that are 'glabrous'. Cassava genotypes have been shown to have a considerable effect on *T. aripo* abundance and, by extension, on its potential to control cassava green mite (Hanna *et al.,* 2007). The predator was also shown to feed on cassava extrafoliar exudates, which have a synergistic positive effect on its reproduction. When *T. aripo* was offered a mixed diet of cassava green mite and exudates, a higher level of biological control of cassava green mite was achieved (R. Hanna *et al.*, unpublished data). Moreover, germplasm surveys showed that hairiness and exudates are positively correlated and that hairiness is a heritable trait that can be bred into cassava clones. This offers a novel perspective to cassava breeding and calls for a paradigm shift: cassava improvement should go beyond traditional breeding traits and include suitability of a clone to natural enemies. The information on the effect of apex hairiness has been widely disseminated and presently numerous cassava improvement programs routinely incorporate the hairiness trait in their clonal selection.

The biological control project against cassava mealybug is essentially finished and the natural enemy cultures are maintained at IITA-Benin only in case the mealybug invades Madagascar or Asia. The project against green mite, however, continues with emphasis on selecting and distributing cassava varieties that are suitable for the predatory mites. Releases of *N. tanajoae* and monitoring of its establishment and impact are still executed in areas where climate matching modelling has suggested a good likelihood of success.

Apart from technical training, higher degree training in collaboration with African, European and American universities was undertaken. Among the dozens of PhD, MSc and hundreds of ing. agr. students trained, many occupy now important positions in their respective countries. Together with assistance to the national programs, this institutional impact enabled many countries to develop south-south collaboration and to tackle new biological control projects, removing their countries at least partially from the grip of powerful insecticide companies. The leader of the cassava mealybug project until 1994, Hans Herren, received the World Food Prize, which brought the project to international attention. Today, biological control is an acknowledged integral part of IPM, recognized as its basis, and there are strong efforts to avoid insecticide abuse. Among the accepted livelihood assets, the projects thus improved the natural capital (avoided pollution), financial assets (documented savings on national bases), human capital (training) and social capitals (continent-wide network of biological control workers). Thus, biological control was (re-) established as an efficient, non-polluting technology with gender-neutral equal impact on poor and wealthy farmers. The costs of the projects were borne by the international donor agencies, which had the opportunity to invest in a long-term, high-interest and corruption-free enterprise.

Do the farmers know this? Throughout the projects, care was taken to inform the farmers wherever possible. Since classical biological control hardly needs any input by farmers, the real clients are the national program and extension officers. In the case of cassava mealybug, the first region-wide official recognition for impact was given at two international conferences in the mid-1990s, i.e., many years after we had quantified impact in these countries in collaboration with our collaborators. Up to today, we receive calls for so-called resurgences and apparent failures of biological control from our colleagues, even from IITA. And in each case, when we follow up, this turns out to be a short, local and relatively small peak in abundance of the pest, representing rather erratic fluctuations around the ten-time lower mean compared to the condition before the introduction of the exotic natural enemies. Infestations of cassava mealybug are, however, higher where farming practices are bad, e.g., under stress conditions when cassava grows on pure sand without mulch. To correct this, better soil conservation practices are needed and the often asked for additional release of the already established natural enemies is only a political palliative. The conclusion is that biological control and functional biodiversity cannot replace good farming.

For Africa this also means that in addition to manual labour, more inputs are needed to replace the nutrients of the depleted soils. To nourish the increasing human populations, we need more productivity on the already farmed land, so that agriculture does not have to expand ever more onto marginal and/or unproductive, but often biodiversity rich land, like the last forests and official nature reserves. The millennium goal of long-term food security and sustainable agricultural production preserving the natural resources, including biodiversity, poses a great challenge, particularly to Africa. Biological control projects like the ones described here are at least moving agriculture in the right direction of sustained food security by increasing agricultural productivity, reducing poverty and protecting the environmental (Neuenschwander *et al.*, 2003). They remain, however, insignificant if the political vision to work towards these goals is not backed by actions, which can be expected to hurt some segments of the population on national, continental, and global scales.

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